

Anti-Jamming 3D Trajectory Design for UAV-Enabled Wireless Sensor Networks Under Probabilistic LoS Channel

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Abstract—This paper investigates the anti-jamming three-dimensional (3D) trajectory design to safeguard the legitimate communications of unmanned aerial vehicle (UAV)-enabled wireless sensor networks (WSNs). Specifically, a UAV is dispatched to collect data over multiple ground sensors (GSs) in the presence of a malicious ground jammer. Under the practical probabilistic line-of-sight (LoS) channel model, we aim to maximize the minimum (average) expected rate among GSs over a finite flight period, by jointly optimizing the GS transmission scheduling, the UAV horizontal and vertical trajectories. To make the formulated non-convex problem tractable, we provide an achievable lower bound for the expected rate, based on which an efficient iterative algorithm is proposed to solve it suboptimally by exploiting the block coordinate descent and successive convex approximation techniques. Numerical results show the significant rate improvement of the proposed joint design and provide new insights into the anti-jamming 3D trajectory design under the probabilistic LoS model, as compared to the conventional two-dimensional (2D) UAV trajectory or under LoS channel models.

Index Terms—3D trajectory design, anti-jamming, probabilistic LoS channel, UAV communications, wireless sensor network.

I. INTRODUCTION

Exploiting unmanned aerial vehicles (UAVs) as aerial data collectors for distributed ground sensors (GSs) is expected to bring significant benefits for future wireless sensor networks (WSNs) [1], such as higher data transmissions, larger service coverage and less energy consumption. Such advantages have urged recent studies on the flexible UAV trajectory design for sensing-data collection in UAV-enabled WSNs, including energy-efficient data collection, flight time minimization, data harvesting throughput maximization and so on [2]–[4]. However, due to their line-of-sight (LoS) dominant links, GS-to-UAV communications are highly vulnerable to severe interference caused

by adversarial ground jammers (GJs), thereby rendering a challenging security threat to UAV-enabled WSNs [5]. Therefore, how to design effective anti-jamming techniques to safeguard UAV-enabled WSNs is an important issue to resolve. In [6], anti-jamming strategies based on reinforcement learning were investigated to improve the static UAV-enabled communications against smart jamming attacks. By taking advantage of the flexible mobility of UAVs, anti-jamming trajectories were designed in [7]–[9] to prevent GJs from degrading the legitimate transmission in UAV-enabled communication systems.

Most of the above works assume that the UAVs fly at fixed altitudes in rural areas, and adopt a simplified LoS channel (LC) model for ground-to-air links. However, such an ideal assumption is inaccurate in practice especially when the UAVs are deployed in urban areas with high and dense buildings or obstacles, since it fails to capture the essential characteristics of the more significant shadowing effect on the signal propagation between the GSs and UAVs [10], [11]. In this case, anti-jamming UAV trajectories obtained under the LC model generally lead to degraded performance in practice.

Motivated by the above, this paper investigates the anti-jamming three-dimensional (3D) trajectory design under the practical elevation angle-dependent probabilistic LoS channel (PLC) model for UAV-enabled WSNs against malicious jamming. Specifically, a UAV is dispatched to sequentially collect data from a set of GSs while mitigating the GJ. We aim to maximize the minimum (average) expected data collection rate from all GSs by jointly optimizing the GS transmission scheduling, UAV horizontal and vertical trajectories over a finite flight period. The formulated problem is non-convex and difficult to be solved optimally as it contains intractable non-convex constraints, binary scheduling variables as well as the highly complicated achievable rate expression with respect to UAV 3D trajectory variables. To overcome such difficulties, we first derive a lower-bound for the expected rate and verify its achievability and accuracy. Then, we propose an efficient iterative algorithm to solve the problem suboptimally by optimizing the GS transmission scheduling, UAV horizontal and vertical trajectories alternately. Since the trajectory optimization subproblems are still non-convex, we further transform the problems into equivalent tractable forms, and then solve them based on the successive convex approximation technique. Simulation results validate that the proposed joint design can efficiently balance the trade-off between the elevation angle and the distance between the UAV and its served GS in the PLC model and achieve significant rate enhancement as compared to the conventional anti-jamming trajectory under the simplified LC model or fixed UAV trajectory. Note that a UAV relay network has been studied in [12], while it differs from this paper in the sense that the UAV flies sufficiently high with path loss and small-scale fading instead of the PLC model considered in this paper.

II. SYSTEM MODEL AND PROBLEM FORMULATION

Consider a UAV-enabled WSN where a UAV is employed as an aerial data collector to fly over $S > 1$ GSs and gather data from them within a given mission duration of T , while a malicious GJ transmits jamming signals with the intention of degrading legitimate UAV communications. In practice, the jamming signals may also be unintentional co-channel interference caused by the transmission of other ground nodes. The set of GSs is denoted as $S = \{1, 2, \dots, S\}$, where $|S| = S$. The horizontal locations of GS s and GJ j are represented by $\mathbf{w}_s = [x_s, y_s]^T \in \mathbb{R}^{2 \times 1}$, $\forall s \in S$ and $\mathbf{w}_j = [x_j, y_j]^T \in \mathbb{R}^{2 \times 1}$, respectively. For ease of exposition, time discretization is applied to partition

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T into I equal-time slots, each with sufficiently small interval $\delta = T/I$. As such, the UAV trajectory can be expressed approximately by a length- I 3D sequence $\{(\mathbf{q}^T[i], h[i])\}_{i=1}^I$, where $\mathbf{q}[i] = [x[i], y[i]]^T \in \mathbb{R}^{2 \times 1}$ and $h[i]$ denote the horizontal and vertical coordinates, respectively. Assuming that the UAV is required to return to its initial location by the end of T for providing a periodical service to GSs. In addition, the UAV's maximum horizontal and vertical flying speeds denoted by $\hat{V}$ and $\tilde{V}$ respectively in meter/second (m/s), as well as its flying altitude, are restricted to the following constraints.¹

$$\|\mathbf{q}[i+1] - \mathbf{q}[i]\| \leq \hat{\Delta}, \quad (1a)$$

$$|h[i+1] - h[i]| \leq \tilde{\Delta}, H_{\min} \leq h[i] \leq H_{\max}, \forall i \in \mathcal{I}, \quad (1b)$$

where $\hat{\Delta} = \delta \hat{V}$ and $\tilde{\Delta} = \delta \tilde{V}$ are the maximum horizontal and vertical flying distances within each time slot $i \in \mathcal{I} \triangleq \{1, \dots, I\}$, respectively, and $H_{\min}$ and $H_{\max}$ are denoted as the minimum and the maximum altitudes that the UAV can reach for any time slot.

To focus on characterizing the performance limit of the considered UAV-enabled WSN in the presence of jamming interference, we assume that the anti-jamming 3D trajectory design is implemented at a central controller in an offline manner (i.e., before the UAV's mission). Moreover, the GJ's location is assumed to be known². We adopt the practical PLC model for the ground-to-air communication links [15]. Specifically, for ground node $m \in \{s, j\}$, $s \in \mathcal{S}$ in time slot i , the LoS probability is expressed as

$$P_m^L[i] = \frac{1}{1 + ae^{(-b[\theta_m[i] - a])}}, \quad (2)$$

where $a > 0$ and $b > 0$ are constants specified by the practical environment, and

$$\theta_m[i] = \frac{180}{\pi} \arctan\left(\frac{h[i]}{\|\mathbf{q}[i] - \mathbf{w}_m\|}\right) \quad (3)$$

is the elevation angle from ground node m to the UAV in this time slot. The corresponding non-LoS (NLoS) probability can then be obtained as $P_m^N[i] = 1 - P_m^L[i]$.

Assume that each GS transmits data with a constant power p_s only when being waken up for transmission by the UAV³. Accordingly, we define a binary variable $\alpha_s[i] \in \{0, 1\}$, which indicates whether GS s is scheduled for transmitting to the UAV in time slot i or not, i.e., GS s transmits if $\alpha_s[i] = 1$ and keeps silent otherwise. Assuming that only one GS is allowed to transmit data to the UAV in each time slot, we have the following scheduling constraints:

$$\sum_{s=1}^S \alpha_s[i] \leq 1, \forall i \in \mathcal{I}, \quad (4a)$$

$$\alpha_s[i] \in \{0, 1\}, \forall s, i. \quad (4b)$$

According to the PLC model, the data collection rate in bits/second/Hertz (bps/Hz) from the scheduled GS s to the UAV in

¹We ignore the UAV acceleration/deceleration constraint in this paper, which is practically reasonable for scenarios when the UAV can complete its maneuvers within a small portion of its whole mission duration [13].

²The location of the GJ can be obtained beforehand with high accuracy based on existing jammer localization techniques, such as jamming signal strength (JSS)-based localization methods proposed in [14].

³Note that the assumption is reasonable, since given any variables $\mathbf{A}$, $\mathbf{Q}$ and $\mathbf{H}$, the following problem (P0) can be reformulated as a convex optimization problem with respect to each GS's transmit power variables. To obtain a higher achievable rate, each GS has to transmit with a constant power, i.e., its peak power, over the whole flight duration.

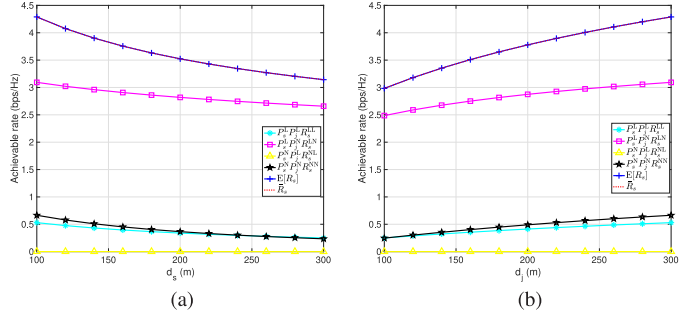


Fig. 1. An illustrative example for the achievability of $\bar{R}_s[i]$. (a) Achievable rate versus d_s . (b) Achievable rate versus d_j .

each time slot is given by

$$R_s[i] = \log_2 \left(1 + \frac{g_s^l[i] p_s}{\sigma^2 + g_j^l[i] p_j} \right), \quad (5)$$

where σ^2 is the receiver noise power and p_j is the maximum transmit power of the GJ. Furthermore, to capture the dominant path loss and shadowing effect on the achieved rate, we model the channel power gain between ground node m and the UAV in each time slot i conditioned on the LoS or NLoS state as $g_m^l[i]$ for $l \in \{L, N\}$ and $m \in \{s, j\}$ [11], where $g_m^L[i] = \beta_0 d_m^{-\alpha_L}[i]$, $g_m^N[i] = \mu \beta_0 d_m^{-\alpha_N}[i]$, $\mu < 1$ is the additional signal attenuation factor owing to the NLoS transmission, β_0 denotes the average channel power at a reference distance $d_0 = 1$ m, $d_m[i] = (\|\mathbf{q}[i] - \mathbf{w}_m\|^2 + h^2[i])^{1/2}$ is the distance between the UAV and ground node m , α_L and α_N represent respectively the average path loss exponents for the LoS and NLoS states.

In a statistical sense, the expected rate under the PLC model from GS s to the UAV in time slot i is given by

$$\mathbb{E}[R_s[i]] = P_s^L[i] P_j^L[i] R_s^{LL}[i] + P_s^L[i] P_j^N[i] R_s^{LN}[i] + P_s^N[i] P_j^L[i] R_s^{NL}[i] + P_s^N[i] P_j^N[i] R_s^{NN}[i], \quad (6)$$

where

$$R_s^{LL}[i] = \log_2 \left(1 + \frac{\gamma_s d_s^{-\alpha_L}[i]}{1 + \gamma_j d_j^{-\alpha_L}[i]} \right), \quad (7)$$

$$R_s^{LN}[i] = \log_2 \left(1 + \frac{\gamma_s d_s^{-\alpha_L}[i]}{1 + \gamma_j \mu d_j^{-\alpha_N}[i]} \right), \quad (8)$$

$$R_s^{NL}[i] = \log_2 \left(1 + \frac{\gamma_s \mu d_s^{-\alpha_N}[i]}{1 + \gamma_j d_j^{-\alpha_L}[i]} \right), \quad (9)$$

$$R_s^{NN}[i] = \log_2 \left(1 + \frac{\gamma_s \mu d_s^{-\alpha_N}[i]}{1 + \gamma_j \mu d_j^{-\alpha_N}[i]} \right), \quad (10)$$

denote respectively the achievable rates at the UAV conditioned on the LoS and NLoS states of the ground-to-air channel, and $\gamma_m = \frac{\beta_0 p_m}{\sigma^2}$. Thus, the expected rate in (6) can be approximated by its lower bound as

$$\mathbb{E}[R_s[i]] \geq P_s^L[i] P_j^L[i] R_s^{LL}[i] + P_s^L[i] P_j^N[i] R_s^{LN}[i] + P_s^N[i] P_j^N[i] R_s^{NN}[i] \triangleq \bar{R}_s[i]. \quad (11)$$

To demonstrate the accuracy and achievability of $\bar{R}_s[i]$, we provide an illustrative example as shown in Fig. 1. The simulation parameters are set as: $p_s = 0.1$ W, $p_j = 0.1$ W, $\beta_0 = -60$ dB, $\sigma^2 = -110$ dBm,

$\alpha_L = 2.2$, $\alpha_N = 3.2$, and $\mu = -20$ dB. For a typical LoS probability in time slot i , e.g., $P_s^L[i] = 0.5$ and $P_j^L[i] = 0.5$, it is observed that the approximated lower bound in (11) is identical with the expected rate in (6) and thus is achievable.

We aim to maximize the minimum average data collection rate among all GSs over I , and the corresponding optimization problem can be formulated as

$$\begin{aligned} \text{(P0)} : \quad & \max_{\mathbf{A}, \mathbf{Q}, \mathbf{H}, \Theta_m, \eta} \eta \\ \text{s.t.} \quad & \frac{1}{I} \sum_{i=1}^I \alpha_s[i] \bar{R}_s[i] \geq \eta, \forall s, \\ & (\mathbf{q}^T[1], h[1]) = (\mathbf{q}^T[I], h[I]), \\ & (1), (3), (4), \end{aligned} \quad (12a)$$

$$(12b)$$

where $\mathbf{A} = \{\alpha_s[i], \forall s, i\}_{i=1}^I$, $\mathbf{Q} = \{\mathbf{q}[i], \forall i\}_{i=1}^I$, $\mathbf{H} = \{h[i], \forall i\}_{i=1}^I$ and $\Theta_m = \{\theta_m[i], \forall i, m \in \{s, j\}\}_{i=1}^I$. (P0) is difficult to be solved optimally because: 1) $\bar{R}_s[i]$ in (11) is non-concave with respect to UAV 3D trajectory variables, due to the non-concave LoS probability of the ground-to-UAV channel given in (2) as well as the non-concave achievable rates shown in (7), (8) and (10); 2) constraints (12a) are non-convex, and the 3D trajectory and GS scheduling variables are coupled; and 3) it contains the non-convex binary constraints (4b) for GS transmission scheduling and non-affine constraints (3) owing to the elevation angle. In fact, (P0) is a mixed-integer non-linear program (MINLP) problem and there are generally no efficient methods for solving it optimally.

III. ANTI-JAMMING UAV 3D TRAJECTORY DESIGN

In this section, we propose an efficient iterative algorithm to solve (P0) based on the block coordinate descent and successive convex approximation techniques. To be specific, the GS transmission scheduling, UAV horizontal and vertical trajectory variables are alternately optimized in each iteration until the algorithm converges.

A. GS Transmission Scheduling Optimization

With any given feasible UAV 3D trajectory $\{\mathbf{Q}, \mathbf{H}\}$, this subproblem can be solved by relaxing the binary variables $\alpha_s[i] \in \{0, 1\}$ to continuous variables $0 \leq \alpha_s[i] \leq 1$, which yields the following relaxed problem

$$\begin{aligned} \text{(P1)} : \quad & \max_{\mathbf{A}, \eta} \eta \\ \text{s.t.} \quad & 0 \leq \alpha_s[i] \leq 1, \forall s, i, \\ & (4a), (12a). \end{aligned} \quad (13a)$$

Since (P1) is a standard linear program, it can be solved efficiently by existing solvers, e.g., CVX [16].

B. Anti-Jamming Horizontal Trajectory Optimization

Given any feasible GS scheduling $\mathbf{A}$ and UAV vertical trajectory $\mathbf{H}$, (P0) reduces to the following UAV horizontal trajectory optimization problem

$$\begin{aligned} \text{(P2)} : \quad & \max_{\mathbf{Q}, \Theta_m, \eta} \eta \\ \text{s.t.} \quad & (1a), (3), (12a), (12b). \end{aligned}$$

To tackle the highly complicated non-convex constraints (12a), the slack variables $\mathbf{x}_m = \{x_m[i], \forall i \in \mathcal{I}, m \in \{s, j\}\}$, $\mathbf{z}_m =$

$\{z_m[i], \forall i \in \mathcal{I}, m \in \{s, j\}\}$, $\mathbf{y}_{s,l} = \{y_{s,l}[i], \forall i \in \mathcal{I}, l \in \{L, N\}\}$, and $\mathbf{y}_{j,l} = \{y_{j,l}[i], \forall i \in \mathcal{I}, l \in \{L, N\}\}$ are first introduced into the rate function (11). Then, $\bar{R}_s[i]$ can be rewritten as

$$\begin{aligned} \bar{R}_s[i] = & \frac{1}{x_s[i]x_j[i]} \log_2 \left(1 + \frac{\gamma_s y_{s,L}^{-1}[i]}{1 + \gamma_j y_{j,L}^{-1}[i]} \right) \\ & + \frac{1}{x_s[i]z_j[i]} \log_2 \left(1 + \frac{\gamma_s y_{s,L}^{-1}[i]}{1 + \gamma_j \mu y_{j,N}^{-1}[i]} \right) \\ & + \frac{1}{z_s[i]z_j[i]} \log_2 \left(1 + \frac{\gamma_s \mu y_{s,N}^{-1}[i]}{1 + \gamma_j \mu y_{j,N}^{-1}[i]} \right), \end{aligned} \quad (15)$$

with the following additional constraints:

$$x_m[i] \geq 1 + ae^{(-b[\theta_m[i]-a])}, \quad (16)$$

$$z_m[i] \geq 1 + \frac{1}{a} e^{(b[\phi_m[i]-a])}, \quad (17)$$

$$y_{s,l}[i] \geq (\|\mathbf{q}[i] - \mathbf{w}_s\|^2 + h^2[i])^{\alpha_l/2}, \quad (18)$$

$$y_{j,l}[i] \leq (\|\mathbf{q}[i] - \mathbf{w}_j\|^2 + h^2[i])^{\alpha_l/2}, \quad (19)$$

where

$$\theta_m[i] \leq \frac{180}{\pi} \arctan \left(\frac{h[i]}{\|\mathbf{q}[i] - \mathbf{w}_m\|} \right), \quad (20)$$

$$\phi_m[i] \geq \frac{180}{\pi} \arctan \left(\frac{h[i]}{\|\mathbf{q}[i] - \mathbf{w}_m\|} \right), \quad (21)$$

are the relaxed constraints for the sake of handling the non-affine constraints (3). We can prove by contradiction that constraints (16)-(21) must hold with equalities that are also met with $\phi_m[i] = \theta_m[i]$ to guarantee the non-decrease of the objective value of (P2).

Despite the introduction of slack variables and the relaxation of non-affine constraints, (P2) is still non-convex for which the optimal solution is challenging to obtain. To solve this problem, we transform $\bar{R}_s[i]$ as below,

$$\bar{R}_s[i] = \bar{R}_s^+[i] - \bar{R}_s^-[i], \quad (22)$$

where $\bar{R}_s^+[i] = \frac{1}{x_s[i]x_j[i]} \log_2(1 + \frac{\gamma_s}{y_{s,L}[i]} + \frac{\gamma_j}{y_{j,L}[i]}) + \frac{1}{x_s[i]z_j[i]} \log_2(1 + \frac{\gamma_s}{y_{s,L}[i]} + \frac{\gamma_j \mu}{y_{j,N}[i]}) + \frac{1}{z_s[i]z_j[i]} \log_2(1 + \frac{\gamma_s \mu}{y_{s,N}[i]} + \frac{\gamma_j \mu}{y_{j,N}[i]})$ and $\bar{R}_s^-[i] = \frac{1}{x_s[i]x_j[i]} \log_2(1 + \frac{\gamma_j}{y_{j,L}[i]}) + \frac{1}{x_s[i]z_j[i]} \log_2(1 + \frac{\gamma_j \mu}{y_{j,N}[i]}) + \frac{1}{z_s[i]z_j[i]} \log_2(1 + \frac{\gamma_j \mu}{y_{j,N}[i]})$.

Since given any $A > 0$ and $B > 0$, the function $f(x_1, x_2, y_1, y_2) = \frac{1}{x_1} \frac{1}{x_2} \log_2(1 + \frac{A}{y_1} + \frac{B}{y_2})$ is jointly convex with respect to its positive variables, we can prove that all terms in $\bar{R}_s^+[i]$ are convex with respect to their corresponding variables. Thus, by applying the first-order Taylor expansion at any given point, $\bar{R}_s^+[i]$ can be approximated by its global lower bound as below.

Lemma 1: For any given UAV horizontal trajectory $\mathbf{Q}^k = \{\mathbf{q}^k[i], \forall i\}$ in the k -th iteration, $\bar{R}_s^+[i]$ given in (22) can be lower-bounded by

$$\begin{aligned} \bar{R}_s^+[i] \geq & f_1^k + f_2^k + f_3^k + (f_{1,x_s[i]}^k + f_{2,x_s[i]}^k)X_s[i] + f_{1,x_j[i]}^k X_j[i] \\ & + f_{3,z_s[i]}^k Z_s[i] + (f_{2,z_j[i]}^k + f_{3,z_j[i]}^k)Z_j[i] + f_{3,y_{s,N}[i]}^k Y_{s,N}[i] \\ & + (f_{1,y_{s,L}[i]}^k + f_{2,y_{s,L}[i]}^k)Y_{s,L}[i] + f_{1,y_{j,L}[i]}^k Y_{j,L}[i] \\ & + (f_{2,y_{j,N}[i]}^k + f_{3,y_{j,N}[i]}^k)Y_{j,N}[i] \triangleq \bar{R}_s^{+,lb}[i], \end{aligned} \quad (23)$$

where $X_m[i] = x_m[i] - x_m^k[i]$, $Z_m[i] = z_m[i] - z_m^k[i]$, $Y_{m,l}[i] = y_{m,l}[i] - y_{m,l}^k[i]$, $m \in \{s, j\}$, $l \in \{L, N\}$ and all other constant coefficients are defined in Appendix A.

Proof: Please see Appendix A. ■

To further handle the complicated terms in $\bar{R}_s[i]$, $\bar{R}_s[i]$ can be equivalently expressed as

$$\bar{R}_s[i] = \bar{R}_s^{+,lb}[i] - \Omega_{1,s}[i] - \Omega_{2,s}[i] - \Omega_{3,s}[i], \quad (24)$$

with additional constraints

$$\begin{cases} \Omega_{1,s}[i] \geq \frac{1}{x_s[i]x_j[i]} \Lambda_1[i], & \Lambda_1[i] \geq \log_2 \left(1 + \frac{\gamma_j}{y_{j,L}[i]} \right), \\ \Omega_{2,s}[i] \geq \frac{1}{x_s[i]z_j[i]} \Lambda_2[i], & \Lambda_2[i] \geq \log_2 \left(1 + \frac{\gamma_j \mu}{y_{j,N}[i]} \right), \\ \Omega_{3,s}[i] \geq \frac{1}{z_s[i]z_j[i]} \Lambda_2[i], \end{cases} \quad (25)$$

where the equality must hold since otherwise $\bar{R}_s[i]$ will decrease. Thus, for all positive variables in (25), the above constraints can be reformulated as the following inequalities based on the first-order Taylor expansion at local points $\Lambda^k = \{\Lambda_1^k[i], \Lambda_2^k[i]\}_{i=1}^I$ in the k -th iteration,

$$\begin{cases} \log_2(x_s[i]) + \log_2(x_j[i]) + \log_2(\Omega_{1,s}[i]) \\ \geq \log_2(\Lambda_1^k[i]) + \frac{\Lambda_1[i] - \Lambda_1^k[i]}{\Lambda_1^k[i] \ln 2}, \Lambda_1[i] \geq \log_2(1 + e^{v_1[i]}), \\ \log_2(x_s[i]) + \log_2(z_j[i]) + \log_2(\Omega_{2,s}[i]) \\ \geq \log_2(\Lambda_2^k[i]) + \frac{\Lambda_2[i] - \Lambda_2^k[i]}{\Lambda_2^k[i] \ln 2}, \Lambda_2[i] \geq \log_2(1 + e^{v_2[i]}), \\ \log_2(z_s[i]) + \log_2(z_j[i]) + \log_2(\Omega_{3,s}[i]) \\ \geq \log_2(\Lambda_2^k[i]) + \frac{\Lambda_2[i] - \Lambda_2^k[i]}{\Lambda_2^k[i] \ln 2}, \end{cases} \quad (26)$$

where

$$y_{j,L}[i] \geq \gamma_j e^{-v_1[i]}, y_{j,N}[i] \geq \gamma_j \mu e^{-v_2[i]}. \quad (27)$$

It can be observed that constraints (26) and (27) are both convex now.

Note that although constraints (19) and (20) are non-convex, the functions in the right-hand-side (RHS) of (19) and (20) are convex with respect to $\mathbf{q}[i]$ and $\|\mathbf{q}[i] - \mathbf{w}_m\|$, respectively. As such, the first-order Taylor expansion at any local point $\mathbf{Q}^k$ can also be used to obtain the following constraints,

$$y_{j,l}[i] \leq (d_j^k[i])^{\alpha_l} + \alpha_l (d_j^k[i])^{\alpha_l - 2} (\mathbf{q}^k[i] - \mathbf{w}_j)^T (\mathbf{q}[i] - \mathbf{q}^k[i]), \quad (28)$$

$$\begin{aligned} \theta_m[i] &\leq \frac{180}{\pi} \arctan \left(\frac{h[i]}{\|\mathbf{q}[i] - \mathbf{w}_m\|} \right) \\ &= \frac{180}{\pi} (F_m^k[i] - G_m^k[i] (\|\mathbf{q}[i] - \mathbf{w}_m\| - \|\mathbf{q}^k[i] - \mathbf{w}_m\|)), \end{aligned} \quad (29)$$

where $d_j^k[i] = (\|\mathbf{q}^k[i] - \mathbf{w}_j\|^2 + h^2[i])^{1/2}$, $F_m^k[i] = \arctan(\frac{h[i]}{\|\mathbf{q}^k[i] - \mathbf{w}_m\|})$, and $G_m^k[i] = \frac{h[i]}{\|\mathbf{q}^k[i] - \mathbf{w}_m\|^2 + h^2[i]}$.

With (26)–(29), (P2) can be reformulated into the following convex optimization problem,

$$\begin{aligned} (P3) : \quad & \max_{\mathbf{Q}, \Theta_m, \Phi_m, \Lambda, \Omega, \mathbf{x}_m, \mathbf{z}_m, \mathbf{y}_{s,l}, \mathbf{y}_{j,l}, \mathbf{v}, \eta} \eta \\ \text{s.t.} \quad & \frac{1}{I} \sum_{i=1}^I \alpha_s[n] (\bar{R}_s^{+,lb}[i] - \Omega_{1,s}[i] - \Omega_{2,s}[i] - \Omega_{3,s}[i]) \geq \eta, \forall s, \\ & (1a), (12b), (16) - (18), (21), (26) - (29). \end{aligned}$$

where $\Phi_m = \{\phi_m[i]\}_{i=1}^I$, $\Lambda = \{\Lambda_1[i], \Lambda_2[i]\}_{i=1}^I$, $\Omega = \{\Omega_{1,s}[i], \Omega_{2,s}[i], \Omega_{3,s}[i], \forall s\}_{i=1}^I$, and $\mathbf{v} = \{v_1[i], v_2[i]\}_{i=1}^I$. As such, (P3) can be efficiently solved by CVX. Note that due to the successive convex approximation in each iteration as shown in (26), (28) and (29), solving (P3) always provides a lower bound of the optimal objective value of (P2).

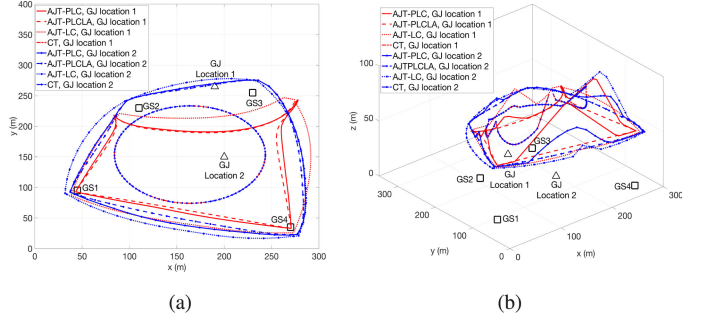


Fig. 2. UAV trajectories by different schemes for different GJ locations. (a) UAV horizontal trajectories (b) 3D UAV trajectories.

C. Anti-Jamming Vertical Trajectory Optimization

Last, given the GS scheduling $\mathbf{A}$ and UAV horizontal trajectory $\mathbf{Q}$ obtained by solving (P1) and (P3), (P0) can be reformulated as the following UAV vertical trajectory optimization problem

$$(P4) : \max_{\mathbf{H}, \Theta_m, \eta} \eta$$

$$\text{s.t. } (1b), (3), (12a), (12b).$$

Note that to solve the non-convex (P4), a similar approach (by using slack variables and SCA techniques) for tackling (P2) can also be applied due to similar forms of the two problems, with minor difference in terms of optimization variables $\mathbf{H}$. Then a lower bound for the objective value of (P4) can be obtained by approximating the non-convex constraints. We omit the detailed procedure owing to the space limitation.

In summary, a suboptimal solution can be efficiently obtained by solving the three subproblems alternately, until the fractional increase of the objective value converges to a pre-set precision, $\epsilon > 0$. Note that by applying the reconstruction method proposed in [17], the binary solution can be reconstructed from the continuous GS transmission scheduling with high accuracy. Moreover, the proposed algorithm is guaranteed to converge over iterations and its total computational complexity is in the order of $O((11N + 9SN)^{3.5} \log^2(1/\epsilon))$.

IV. NUMERICAL RESULTS

In this section, we provide simulation results to validate the effectiveness of the optimized anti-jamming 3D trajectory under the PLC model (denoted by AJT-PLC). For comparison, we consider the following schemes: 1) anti-jamming horizontal trajectory under the PLC model with the lowest altitude being $h[i] = H_{\min}$, $\forall i$ (denoted by AJT-PLCLA); 2) anti-jamming 3D trajectory under the simplified LC model (denoted by AJT-LC), i.e., setting the LoS probability of the ground-to-UAV channel given in (2) to be $P_m^L[i] = 1$, $\forall i, m \in \{s, j\}$; and 3) circular trajectory (denoted by CT) with the circle center as $\mathbf{c} = \frac{\sum_{s=1}^S \mathbf{w}_s}{S}$

and the corresponding radius as $r_{\text{trj}} = \min(\frac{\hat{V}N}{2\pi}, \frac{\max_{s \in S} \|\mathbf{w}_s - \mathbf{c}\|}{2})$ [17], and the GS scheduling as well as the UAV altitude is alternately optimized based on Section III-1) and 3) until convergence. The required simulation parameters are set as follows [11], [15]: $S = 4$, $p_s = 0.1$ W, $p_j = 0.1$ W, $\delta = 0.2$ s, $\hat{V} = 40$ m/s, $\bar{V} = 20$ m/s, $a = 11.95$, $b = 0.14$, $\beta_0 = -60$ dB, $\sigma^2 = -110$ dBm, $\alpha_L = 2.2$, $\alpha_N = 3.2$, $\mu = -20$ dB, and $\epsilon = 10^{-4}$.

Fig. 2, illustrates the UAV trajectories by different schemes with different GJ locations when $T = 40$ s. It is observed in Fig. 2(a) that

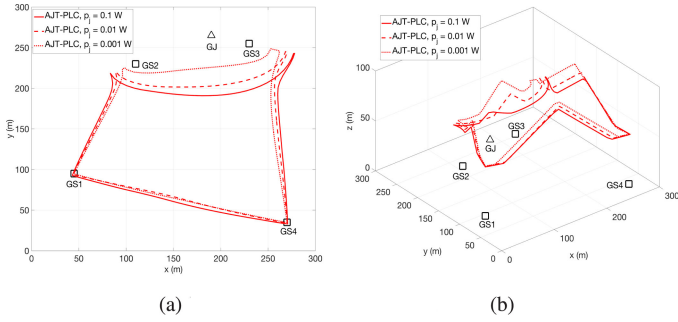


Fig. 3. UAV trajectories for different p_j . (a) UAV horizontal trajectories. (b) 3D UAV trajectories.

in contrast to the AJT-LC scheme, the UAV in the case of AJT-PLC and AJT-PLCLA has the similar horizontal trajectories and it hovers farther away from GSs 2 and 3, but closer to GSs 1 and 4 when the GJ locates at Location 1. This is because when the UAV receives data from GSs 2 and 3, its elevation angles with these two GSs as well as the GJ are relatively larger. Thus, without incurring much rate loss, the UAV trajectory can be adjusted to reduce the elevation angle between the UAV and GJ, which leads to lower LoS probability of the GJ-to-UAV channel. By contrast, since GSs 1 and 4 are farther away from the GJ, the elevation angle between the UAV and GJ is smaller when the UAV receives data from them. Therefore, the UAV trajectory can be set to enlarge the UAV elevation angles with GSs 1 and 4 for further rate improvement. Note that above-mentioned trade-off between the elevation angles and distances under the practical PLC model cannot be observed in the UAV trajectories by AJT-LC. For the UAV 3D trajectories as shown in Fig. 2(b), the proposed AJT-PLC scheme can take advantage of the additional design of the UAV vertical trajectory to obtain the more effective angle-distance trade-off than AJT-PLCLA since the elevation angle can be adaptively adjusted according to the locations of its associated GS and the GJ. It is also worth mentioning that although the trajectories of AJT-PLC and AJT-LC seem similar at some locations when the GJ is located at Location 2 (e.g., geometric center of GSs), the achieved max-min rates of the two schemes are significantly different. Since the rate performance for the typical locations of the GJ results in similar observations, we only show the results for the GJ at Location 1 due to the space limitation.

Fig. 3, compares the UAV trajectories by the proposed AJT-PLC scheme for different p_j when $T = 40$ s. First, as shown in Fig. 3(a), when the GJ transmits with a lower jamming power, the UAV can adjust its trajectory to move closer to each GS, since the UAV can achieve a larger elevation angle with each GS when hovering closely above each GS. As such, the UAV suffers the lower path loss and attains more transmission rate with each GS. Second, it is expected that the UAV hovers close to each GS at its lowest altitude for a certain amount of time. This is because at this location, the UAV can strike an optimal balance between the larger elevation angle and the smaller path loss, and thus achieving higher rate. Finally, with the increase of p_j , the UAV will lower its flying altitude when traveling between the GSs since this helps decrease its elevation angle with the GJ to prevent from further interfering the UAV due to the higher LoS probability of the GJ-to-UAV channel.

In Fig. 4(a), we compare the achieved average max-min rate of the UAV by different trajectory designs with different p_j under different T . First, when the UAV suffers from strong interference, e.g., $p_j = 0.1$ W, it is observed that despite of no anti-jamming trajectory design, the CT scheme still outperforms the AJT-LC scheme significantly, owing

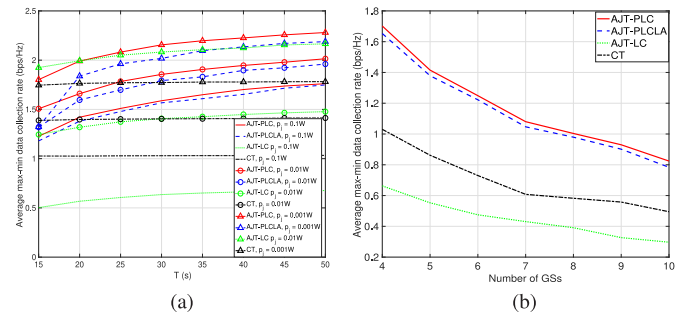


Fig. 4. Average max-min rate by different trajectory designs. (a) Max-min rate versus T . (b) Max-min rate versus number of GSs.

to the more accurate PLC model rather than the simplified LC model. Furthermore, the AJT-PLCLA scheme achieves larger rate than the CT scheme, since it fully exploits the UAV's horizontal trajectory to avoid severe interference. By additionally designing the vertical trajectory, the proposed AJT-PLC outperforms the AJT-PLCLA scheme. This demonstrates the importance of the anti-jamming 3D trajectory design for UAV-enabled WSNs with malicious jammers. Second, although increasing the jamming power leads to lower max-min rates for all schemes, the performance gaps between AJT-PLC and AJT-LC as well as CT become larger, which further demonstrates the significance of anti-jamming 3D trajectory design under the more accurate PLC, especially for the case with more severe jamming. Third, as p_j decreases, the rate of the AJT-LC scheme improves significantly, since in this case the trajectories of AJT-LC and AJT-PLCLA do not differ much when T is sufficiently large. By contrast, the rate gap between AJT-PLCLA and AJT-PLC gradually becomes larger. This is because with lower jamming power, adjusting the UAV altitude is one effective means to further improve the rate performance of AJT-PLC. Finally, Fig. 4(b) illustrates the effects of the number of GSs on the achieved expected max-min rate. The GSs' locations are randomly distributed over a square area of 350×350 m². It is observed that with the number of GSs increasing, the rates of all schemes decrease, as expected. The reason is that the communication resources are shared by more GSs, while the achieved max-min rate of the UAV depends on the worst-case GS with the lowest average rate.

V. CONCLUSION

This paper studied the anti-jamming trajectory 3D design for data collection in UAV-enabled WSNs in the presence of jamming. The GS transmission scheduling and UAV 3D trajectory were jointly optimized to maximize the minimum expected rate of GSs based on the practical PLC model. Then, an efficient iterative algorithm was proposed to obtain a suboptimal solution by leveraging alternating optimization and successive convex approximation techniques. Significant rate performance improvement was shown in the numerical results, which validates the practical significance of considering the more accurate PLC model with anti-jamming 3D trajectory design for UAV-enabled WSNs.

V. APPENDIX

A. Proof of Lemma 1

Since $f(x_1, x_2, y_1, y_2) = \frac{1}{x_1 x_2} \log_2(1 + \frac{A}{y_1} + \frac{B}{y_2})$ (use f for notation simplicity) is jointly convex with respect to their positive variables, based on the first-order Taylor expansion at any given local points x_1^k ,

x_2^k, y_1^k , and y_2^k in the k -th iteration, we have

$$f \geq f^k + f_{x_1}^k X_1 + f_{x_2}^k X_2 + f_{y_1}^k Y_1 + f_{y_2}^k Y_2, \quad (32)$$

where

$$f_{x_1}^k = -\frac{\log_2(C^k)}{(x_1^k)^2 x_2^k}, f_{y_1}^k = -\frac{A \log_2 e}{x_1^k x_2^k (y_1^k)^2 C^k},$$

$$f_{x_2}^k = -\frac{\log_2(C^k)}{x_1^k (x_2^k)^2}, f_{y_2}^k = -\frac{B \log_2 e}{x_1^k x_2^k (y_2^k)^2 C^k},$$

$f^k = \frac{1}{x_1^k x_2^k} \log_2(1 + \frac{A}{y_1^k} + \frac{B}{y_2^k})$, $C^k = 1 + \frac{A}{y_1^k} + \frac{B}{y_2^k}$, $X_m = x_m - x_m^k$ and $Y_m = y_m - y_m^k$, $m \in \{1, 2\}$. Denote by $f_1 = \frac{1}{x_s[i] x_j[i]} \log_2(1 + \frac{\gamma_s}{y_{s,L}[i]} + \frac{\gamma_j}{y_{j,L}[i]})$, $f_2 = \frac{1}{x_s[i] z_j[i]} \log_2(1 + \frac{\gamma_s}{y_{s,L}[i]} + \frac{\gamma_j \mu}{y_{j,N}[i]})$ and $f_3 = \frac{1}{z_s[i] z_j[i]} \log_2(1 + \frac{\gamma_s \mu}{y_{s,N}[i]} + \frac{\gamma_j \mu}{y_{j,N}[i]})$. Due to the similar forms for each term in $\bar{R}_s^+[i]$ of (22), we can complete the proof of Lemma 1 by applying (32).

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